

YÜKSEK TEKNOLOJİK UYGULAMALAR İÇİN ATIK MALZEMELERDEN RUBİDYUM GERİ KAZANIMI

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Özet

Sınırlı bolluğuna rağmen, rubidyum çeşitli bilimsel ve endüstriyel uygulamalar için giderek daha önemli hale gelmektedir. GPS gibi kritik sistemleri destekleyen atomik saatler ve kuantum sensörler de dahil olmak üzere kuantum teknolojilerinde hayati bir rol oynamaktadır. Yüksek teknoloji endüstrilerinde rubidyuma olan talebin artmasıyla birlikte, verimli çıkarım ve sürdürülebilir tedarik yöntemleri üzerine yapılan araştırmalar hız kazanmıştır. İyon değişimi ve solvent ekstraksiyon gibi teknikler, kaynak sürdürülebilirliği zorluklarını ele alarak tuzlu su kaynaklarından rubidyum geri kazanmak için geliştirilmektedir. Bu derleme, rubidyumun çıkarım yöntemleri, teknolojik uygulamaları ve sürdürülebilirlik çabalarındaki son gelişmeleri sentezleyerek, modern bilim ve sanayinin ihtiyaçlarını karşılamadaki artan potansiyelini vurgulamaktadır.

Anahtar Kelimeler: Rubidyum, Geri kazanım, Atık malzemeler, Yüksek teknoloji uygulamaları

RECOVERY OF RUBIDIUM FROM WASTE MATERIALS FOR HIGH-TECH APPLICATIONS

Abstract

Despite its limited abundance, rubidium is increasingly integral to various scientific and industrial applications. It is crucial in quantum technologies, including atomic clocks and quantum sensors, which support critical systems like GPS. With the growing demand for rubidium in high-tech industries, research on efficient extraction and sustainable supply methods has gained momentum. Techniques such as ion exchange and solvent extraction are being refined to recover rubidium from brine sources, addressing resource sustainability challenges. This review synthesizes recent advancements in rubidium's extraction methods, technological applications, and sustainability efforts, emphasizing its expanding potential in addressing the needs of modern science and industry.

Keywords: Rubidium, Recovery, Waste materials, High-Tech applications

1. Introduction

Rubidium (Rb) is a soft, silvery-white metallic element classified within the alkali metal group, characterized by its high reactivity and unique properties. With an atomic number of 37, rubidium was first identified in 1861 by Robert Bunsen and Gustav Kirchhoff during spectroscopic analysis of the mineral lepidolite. Its name, derived from the Latin word *rubidus*-meaning deep red-refers to the striking red spectral lines observed during its discovery. Although it is the second least abundant alkali metal in the Earth's crust, with an average concentration of approximately 90 ppm, rubidium is still more abundant than elements such as lithium, underlying its potential as a valuable resource (Janev et al., 1985; Emsley, 2011).

Despite its relative scarcity, rubidium's distinctive physical and chemical properties have led to diverse applications across multiple scientific and industrial domains. In quantum technology, rubidium atoms play a pivotal role in the creation of atomic clocks and quantum sensors. These technologies underpin critical systems such as global navigation satellite networks, including GPS, where precise timekeeping is essential. The element's utility extends into medicine, where the radioactive isotope Rubidium-82 is employed in positron emission tomography (PET) scans. This isotope is instrumental in evaluating myocardial perfusion, offering invaluable insights into cardiovascular health. Moreover, the increasing demand for rubidium in high-tech industries has spurred research into efficient recovery and sustainable supply strategies. Advanced methods such as ion exchange and solvent extraction are being optimized for rubidium recovery from brine sources, aiming to address challenges in resource sustainability and meet growing industrial requirements (Chen et al., 2020; Bello et al., 2021).

This review aims to provide a comprehensive synthesis of recent research on rubidium, with a focus on its extraction, applications, and advancements in technology and sustainability. By examining its role in critical areas such as quantum science, medical diagnostics, and resource recovery, this study highlights the expanding potential of rubidium in addressing modern scientific and industrial challenges.

2. Waste Materials

2.1. Boron Waste

Ertan and Erdogan (2017) evaluated two methods for extracting rubidium from Emet/Kütahya boron clay wastes. Direct acid leaching with sulfuric acid achieved 43.8% efficiency, favored for its simplicity and cost-effectiveness. The second method, roasting at 950°C with calcium sulfate and carbonate followed by hydrofluoric acid leaching, achieved 87.43% efficiency but posed scalability and safety challenges. Solvent extraction using 18-Crown-6 in nitrobenzene at pH 8 enhanced rubidium recovery, demonstrating its selectivity.

Özbaş and Derun (2021) investigated rubidium recovery from boron waste and borax clay, calcining samples with calcite and calcium chloride at 650–750°C, followed by water leaching. Obut et al. (2020) optimized rubidium leaching from Kırka boron clays, achieving 65.2% efficiency under specific conditions (1-hour leaching, 2 M H₂SO₄, 90°C). Özmal et al. (2021) used potassium cobalt hexacyanoferrate to adsorb rubidium from boron waste solutions pre-treated with hydrochloric and nitric acids.

2.2. Gold Beneficiation Plant Wastes

Mohammadi et al. (2014) developed a three-phase flow sheet for rubidium recovery from gold waste, achieving 97.14% efficiency via impurity removal, salt roasting, and water leaching (Figure 1). In a follow-up study (Mohammadi et al., 2015), the team employed 18-crown-6 in kerosene for solvent extraction, achieving a 99.12% rubidium recovery rate, highlighting the method's efficacy in purifying and concentrating rubidium.

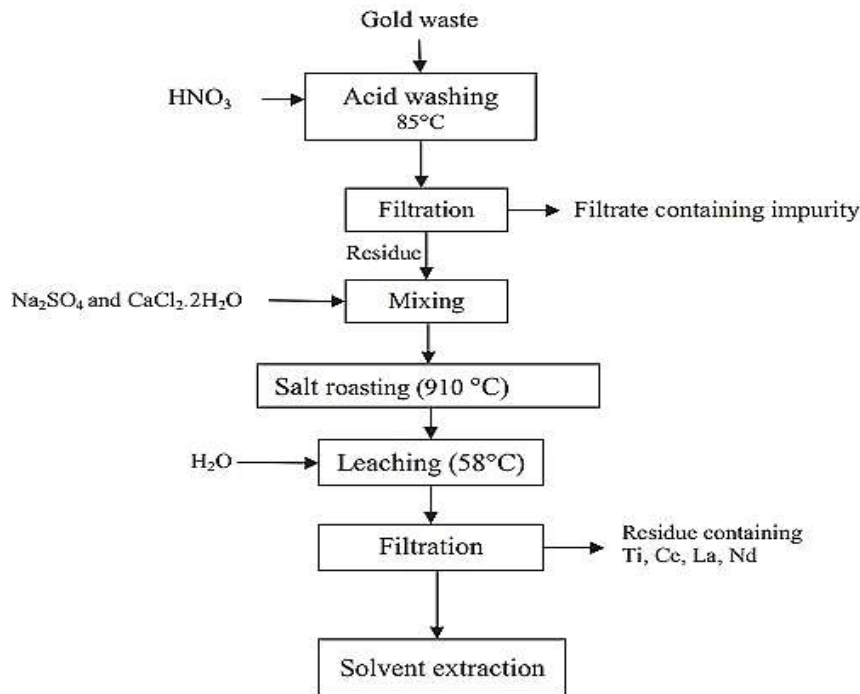


Figure 1. Flow sheet for rubidium recovery from gold waste.

Silver Cobalt Works reported a 99% rubidium recovery rate from Granada Gold Mines in Canada using the Re-2Ox process (Basa, 2024). This acid leaching method is environmentally friendly and offers a sustainable solution for extracting critical elements, particularly for electric vehicle production. The Re-2Ox process represents a significant advancement in sustainable mining technologies, aligning with the increasing demand for green resource recovery. Figure 2 outlines its key steps and operational conditions, highlighting its efficiency and environmental benefits.

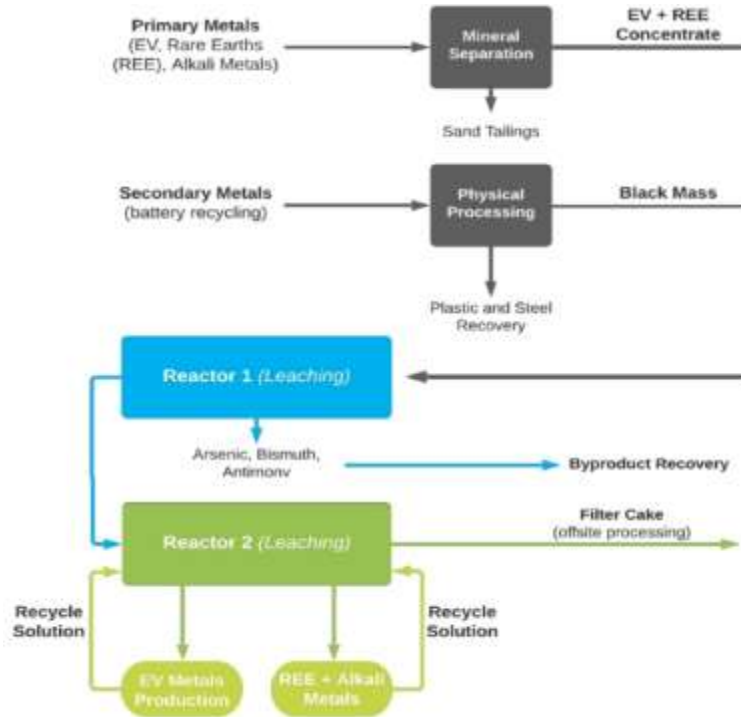


Figure 2. Flow sheet of Re-2Ox process.

2.3. Clay Wastes

Siame and Pascoe (2011) explored the pre-enrichment of lithium and rubidium from clay wastes using flotation and magnetic separation, enhancing metal concentrations for subsequent extraction. This approach demonstrated its effectiveness as a pre-treatment method, contributing to sustainable resource recovery. Similarly, Zhou et al. (2015) developed a two-step process for extracting rubidium from kaolin clay waste, involving chlorination roasting followed by water leaching. This method addresses the growing demand for rubidium in advanced technologies such as quantum devices, medical imaging, and energy storage.

2.4. Brines

Fang et al. (2023) developed a two-step adsorption-flotation method to separate cesium and rubidium from brines. Ammonium molybdophosphate was used for selective ion adsorption, followed by flotation with cetyltrimethylammonium bromide (CTAB) to isolate the adsorbed metals efficiently. Lee et al. (2024) investigated hydrometallurgical methods to recover rubidium from desalination brine, focusing on valorizing brine waste. The process involved brine pretreatment, rubidium adsorption using ion exchange resins, and desorption with acids. The study emphasized the effectiveness of t-BAMBP combined with ionic liquids like C₂mimNTf₂, enabling highly selective and efficient rubidium extraction.

2.5. Fly Ashes

Fly ashes, by-products of coal combustion, are emerging as a viable source of rubidium, alongside rare earth elements (REEs) and platinum group metals (PGMs) (Moreno et al., 2005; Izquierdo et al., 2011). Utilizing fly ash for rubidium recovery not only addresses environmental concerns but also offers an alternative supply of this valuable alkali metal. Nygren (2019) developed a multi-stage process for rubidium recovery from fly ash. The initial oxalic acid leaching achieved a 56% recovery, with rubidium precipitating as oxalate. Subsequent pH adjustment further enhanced recovery efficiency to 62–84% (Figure 3), highlighting the critical role of pH optimization in maximizing rubidium extraction.

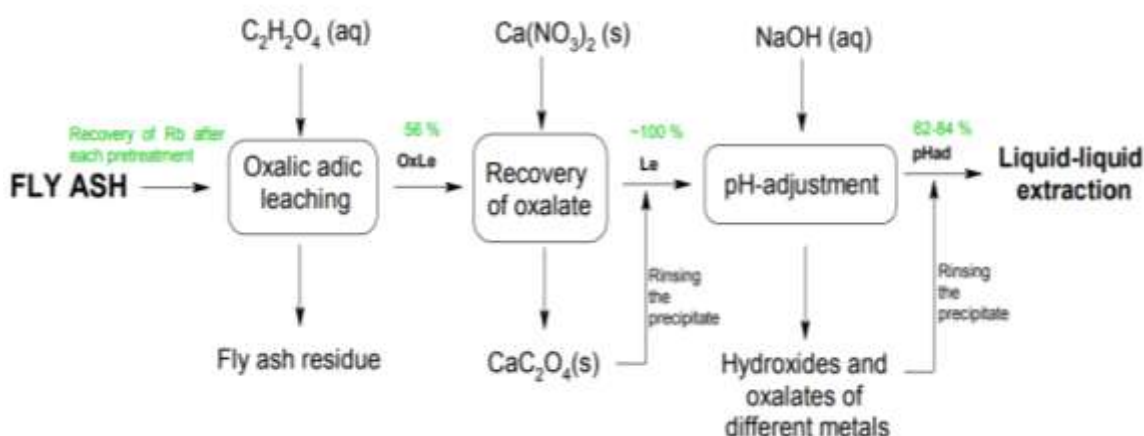


Figure 3. Rubidium recovery from fly ash.

3. Conclusions

Rubidium's role in modern science and technology is both multifaceted and expanding. As a crucial alkali metal, rubidium plays an essential part in various high-tech applications, including quantum computing, medical diagnostics, and energy storage. In quantum computing, rubidium atoms are used in atomic clocks and quantum sensors, essential for advancing precision timekeeping and next-generation technologies. In medical diagnostics, particularly in positron emission tomography scans, the radioactive isotope Rubidium-82 is invaluable for assessing myocardial perfusion, aiding in the diagnosis and management of cardiovascular diseases. The ongoing research into rubidium's extraction methods and its integration into emerging technologies underlines its versatility and growing significance. As scientific exploration continues to unlock new applications, rubidium is expected to become a cornerstone element in the development of future technological innovations. From clean energy solutions to advancements in telecommunications, the increasing demand for rubidium will likely drive further advancements in its extraction, refinement, and utilization, solidifying its position as a critical material in both scientific and industrial fields.

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